

2.1 Expressions and formulae *continued*

Subject content	Guidance	Curriculum reference
P Solve problems in context, involving formulae and expressions	The perimeter, P, of a rectangle can be found using the formula: $P = 2l + 2w$ Where l = length of the rectangle and w = width of the rectangle The perimeter of a rectangle is 156 cm The length of the rectangle is twice as long as its width. What is the length of the rectangle?	A7.1I A8.1F A9.1J

2.2 Equations

Subject content	Guidance	Curriculum reference
A Find pairs of numbers that satisfy a linear equation with two unknowns	$x + y = 7$ Write down 2 possible values for x and y	A8.2A
B Solve linear equations including unknowns on both sides	Solve $4x - 3 = 2x + 7$ $5y + 4 = 7y + 5$	A8.2B A9.2A
C Solve linear equations, including unknowns on both sides and/or include brackets or fractions	Solve $\frac{5(x-3)}{4} = \frac{2x+6}{2}$	A8.2B A8.2C A9.2B
D Solve problems by writing and solving equations	A man said 'My brother is 8 years' older than me, my sister is 5 years' younger than me, and all three of our ages add up to 129'. How old is the man's brother?	A8.2D
E Solve equations involving an x^2 term and a number	Solve $x^2 - 25 = 96$	A9.2C
F Solve quadratic equations of the form $x^2 + bx + c = 0$, by factorising	Solve $x^2 - 7x - 30 = 0$	A9.2D

2.2 Equations *continued*

Subject content	Guidance	Curriculum reference
G Write equations to represent direct proportion	When $x=16$, $y=4$ y is directly proportional to x Write a proportionality equation involving x and y	A9.2E
H Solve a pair of simultaneous linear equations	Solve the simultaneous equations: $2x + 3y = -4$ $4x - y = 6$	A9.2F
I Solve problems involving simultaneous linear equations or direct proportion	4 burgers and 3 cakes costs \$14.80 5 burgers and 2 cakes costs \$15.70 How much would 6 burgers and	A9.2G